

Module 07 [UML+Java Basics]

(for Week 8: 19-23/7/2025)

Topic 1 [UML]

Problem statement: Draw the UML diagram for the following class

```
class Test{
    private: int x;
    protected: int y;
    public: int z;
    public:
        void SetData(int a,int b,int c){
            x=a;y=b;z=c;
        }
        float getAverage(){
            Return (x+y+z)/3.0;
        }
}
```

Topic 2 [UML: Aggregation]

Problem statement: Create two classes Doctor and Patient to store the information of doctors and patients respectively. Now write codes to create association between these two classes by aggregation.

[note: if Patient class is destroyed then Doctor class is still exist]

class Doctor{ //write your code }	class Patient{ //write your code }	int main(){ //write your code }
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Topic 3 [UML: Composition]

Problem statement: Create two classes Doctor and Patient to store the information of doctors and patients respectively. Now write codes to create association between these two classes by composition.

[note: if Doctor class is destroyed then Patient class is also destroyed]

class Doctor{ //write your code }	class Patient{ //write your code }	int main(){ //write your code }
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Topic 4: [java Programming Basics]

Problem Statement: Write java programs to do the following

- Display your name and address in two lines
- Take two integers as inputs and prints the bigger one
- Input a series of 10 numbers into an array ax[] and finds the largest, smallest and average of these numbers. Also write code for searching and sorting.

Topic 5 [java Programming private method]

Problem Statement: Verify that private methods can be used only inside the class but it cannot be used from outside the class.

Topic 6 [java Programming static method/member]

Problem Statement: Verify that static methods can access static method/member but cannot access to non-static method/member.

Topic 7 [java Programming]

Problem Statement: Write a Java Program to initialize radius of a circle using constructor/Setter and calculate area of the circle.